

**AMENDMENT NO. 2 JUNE 2015**  
**TO**  
**IS 9798 : 2013 LOW PRESSURE REGULATORS FOR USE WITH LIQUEFIED**  
**PETROLEUM GAS (LPG) MIXTURES**

*(Second Revision)*

*[Second cover page, Foreword, para 2 (c)]* — Delete and renumber the subsequent SI No.

*(Second cover page, Foreword, para 3)* — Substitute the following for the existing para:

‘In the formulation of this standard considerable assistance has been derived from BS EN 16129 : 2013 ‘Pressure regulators, automatic change-over devices, having maximum regulated pressure of 4 bar , with maximum capacity of 150 kg/h , associated safety devices and adaptors for butane, propane, and their mixtures (superseding EN 12864 : 2001).’

*(Page 1, clause 3.6)* — Insert the following new clauses at the end:

**‘3.7 Type of Inlet Connections**

**3.7.1 Quick Coupling**

A connection system which allows the fitting of a regulator to a cylinder valve without a threaded connection and without use of tools.

**3.7.2 Threaded Inlet**

A connection system which allows the fitting of a regulator to a cylinder valve by means of a threaded connection and/or use of tools for fitment / removal.’

*(Page 4, clause 5.5.2)* — Substitute the following for the existing para:

‘5.5.2 The inlet connection shall also be capable to withstand a minimum pneumatic pressure equivalent to the maximum vapour pressure specified in 3.1 at ambient temperature.’

*(Page 4, clause 5.11)* — Substitute the following for the existing clause:

**‘5.11 Strength of the Connection / Regulator Assembly**

**5.11.1** The fixing of the inlet connection on to the regulator body , whether it is of the threaded or non - threaded type or in one piece, shall withstand the following tests, under the conditions defined in 8.11 (*see* Table 1):

- a) A torque of at least 30 Nm in both directions; and
- b) A tensile strength test of 2 000 N

**5.11.2** The fixing of the outlet connection on to the regulator body, whether it is of the threaded, or non-threaded type or in one piece, shall resist the following tests, under the conditions defined in 8.11 (*see* Table 2):

- a) *For non - threaded hose connection :*
  - 1) A torque of at least 30 Nm in one direction (verification not required for freely rotating connection) ,
  - 2) Bending moment of 10 Nm , and
  - 3) A tensile strength test of 2 000 N.

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### **b) For threaded unions:**

- 1) A torque of at least 30Nm in both directions (verification not required for freely rotating connections) ,
- 2) A bending moment of 10 Nm , and
- 3) A tensile strength test of 2 000 N.

**5.11.3** For freely rotating connections, the torque necessary for the rotation of the connection shall not be greater than 0.5 Nm at the end of all the tests carried out.

**5.11.4** No distortion or breakage shall be evident and the regulators shall comply with the soundness test described in **6** after application of the forces.

## **5.12 Strength of Regulator Assembly when Fitted onto a Cylinder Valve**

**5.12.1** The regulators when installed as indicated in the installation instruction shall resist the following tests under the conditions defined in **8.11** (see Table 3):

- a) A torque in both directions,
  - 1) of at least 20 Nm for non-threaded hose outlet connections (15 Nm for quick coupling connections); and
  - 2) of at least 30 Nm for threaded outlet connections. In addition, regulators with screwed unions intended to be vertically mounted onto the cylinder valve, shall resist a torque of at least 20 Nm in the regulator plane (15 Nm for quick coupling connections).
- b) A bending moment created by a force of 400 N directed upwards and whose application point is at the base of the outlet connection, and
- c) A tensile strength test of 500 N, for quick coupling connections only.

**5.12.2** The mechanical strength required shall be ensured for all the positions of fixing of the regulator (as indicated in the installation instructions) onto the cylinder.

**5.12.3** There shall be no distortion or breakage that could affect the safety of the regulator. The regulator shall comply with the soundness test described in **6** after application of the forces.'

(Page 6, clause **6.1**, line 5) — Substitute 'NTP that is 760 mm of Hg and 20 °C' for 'NTP that is 760 mm of Hg and 0 °C'.

(Page 7, clause **8.9.1**, sentence 2) — Substitute the following for the existing sentence:

'Static (lock up) pressure shall not exceed 4.41 kN/m<sup>2</sup> (45 gf/cm<sup>2</sup> or 450 mm of WC).'

(Page 8, clause **8.11**) — Substitute the following for existing:

## **'8.11 Mechanical Strength of Connections**

### **8.11.1 General**

Tests for mechanical strength shall be carried out using a dynamometric device allowing the measurement of forces to within ± 5 percent accuracy. For the torque test, a system which neutralises bending moments shall be used (if a torque wrench is used it is desirable that this is double handed). The duration of application of the torques, moments and forces shall be 60s each.

### **8.11.2 Regulator Intended to be Directly Connected to a Cylinder Valve**

The points where the device is held and the test values are those shown in Tables 1, 2 and 3.

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**8.11.3** For the tests given in Table 3

- a) Devices with threaded connections, shall be mounted on the valve as indicated in the installation instructions; and
- b) Devices with free rotating quick coupling connection, the torque test of 15 Nm is not required.

(Page 8, Table 1) — Substitute the following for the existing table:

**Table 1 Mechanical Strength Test for Inlet Connections**

(Clauses 5.11.1 and 8.11.2)

| SI No.<br>(1)                                                                                     | Test Diagram<br>(2) | Type of Load<br>(3) | Value<br>(4)     |
|---------------------------------------------------------------------------------------------------|---------------------|---------------------|------------------|
| i)                                                                                                |                     | T<br>F              | 30 Nm<br>2 000 N |
| ii)                                                                                               |                     | T<br>F              | 30 Nm<br>2 000 N |
| key ▼▲▼ Regulator holding points<br>T = torque, F = pulling force, C = inlet connection (see 3.7) |                     |                     |                  |

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(Page 8, Table 2) — Substitute the following for the existing table:

**Table 2 Mechanical strength test for outlet connections**  
(Clauses 5.11.2 and 8.11.2)

| Sl No.<br>(1)                                                                                                               | Test Diagram<br>(2) | Type of Load<br>(3) | Value<br>(4)              |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|---------------------------|
| i)                                                                                                                          |                     | T<br>F<br>M         | 30 Nm<br>2 000 N<br>10 Nm |
| ii)                                                                                                                         |                     |                     |                           |
| iii)                                                                                                                        |                     | T<br>F<br>M         | 30 Nm<br>2 000 N<br>10 Nm |
| iv)                                                                                                                         |                     |                     |                           |
| <b>key</b><br>Regulator holding points T = torque, F = pulling force, M = bending moment,<br>C = inlet connection (see 3.7) |                     |                     |                           |

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(Page 8, Table 3) — Substitute the following for the existing table:

**Table 3 Mechanical Strength Test for the Device Assembly mounted on the Cylinder Valve**  
(Clauses 5.12.1, 8.11.2 and 8.11.3)

| Sl No.                                                                                                                                                                                                                                                                                   | Test Diagram | Type of Load                                            | Value for Threaded Inlet Connection | Value for Quick Coupling Connection |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------|-------------------------------------|-------------------------------------|
| (1)                                                                                                                                                                                                                                                                                      | (2)          | (3)                                                     | (4)                                 | (5)                                 |
| i)                                                                                                                                                                                                                                                                                       |              | T<br>F<br>F <sub>1</sub>                                | 20 Nm<br>—<br>400 N                 | 15 Nm<br>500 N<br>400 N             |
| ii)                                                                                                                                                                                                                                                                                      |              | T<br>F<br>F <sub>1</sub>                                | 20 Nm<br>—<br>400 N                 | 15 Nm<br>500 N<br>400 N             |
| iii)                                                                                                                                                                                                                                                                                     |              | T<br>F<br>F <sub>1</sub>                                | 30 Nm<br>—<br>400 N                 | 30 Nm<br>500 N<br>400 N             |
| iv)                                                                                                                                                                                                                                                                                      |              | T <sub>1</sub><br>T <sub>2</sub><br>F<br>F <sub>1</sub> | 30 Nm<br>20 Nm<br>—<br>400 N        | 30 Nm<br>15 Nm<br>500 N<br>400 N    |
| <p>key  Regulator fixing points      F = pulling force<br/>  Valve      F<sub>1</sub> = bending force      C = Inlet connection (see 3.7)<br/> T T<sub>1</sub> T<sub>2</sub> = Torque (T and T<sub>2</sub> are not applied if the regulator is freely turning on its cylinder valve)</p> |              |                                                         |                                     |                                     |